

Trim & Flight Model

Trimming — teaching the glider's control model to match how the vehicle actually flies — is where most of a Seaglider pilot's early-mission effort goes. Pitch trim is arguably the **single biggest influence on flight**: a well-trimmed vehicle flies cleanly on both profiles, dives and climbs in about the same time, and wastes far less energy on the VBD; a badly trimmed one burns battery, and sometimes doesn't fly at all. This page covers the practical trimming workflow (pitch, then buoyancy, then roll), pre-deployment ballasting, and the basestation's automated Flight Model System (FMS).

Source

Paraphrased from the APL-UW IOP office-hours series (pitch trimming; ballasting and volmax; flight-model questions) and webinar series (*Piloting with basestation3*), the UW *Determining Seaglider Velocities Automatically* flight-model paper (Bennett, Stahr & Eriksen), and the APL-UW *SGX Documentation*. Numbers quoted here are IOP starting points, not gospel — confirm parameter definitions in the [Parameter Reference Manual](#) and defer to APL-UW IOP guidance.

Ballasting before the mission: estimating volmax in a tank

Before a sea trial, the glider needs lead trimmed so that its maximum volume (volmax) gives the right thrust in the density of the operating area. IOP's tank procedure:

1. Weigh the glider in air to get its mass M .
2. Soak it **overnight**, fully submerged, in a freshwater tank.
3. Compute the tank water density ρ_{tank} from temperature (and salinity if a saltwater tank).
4. With an accurate scale, measure the glider's weight in water W_i at an intermediate VBD position VBD_i (e.g. 2000 A/D counts).
5. Compute volmax:

$$\text{volmax} = (M - W_i) / \rho_{\text{tank}} + (\$VBD_MIN - \text{VBD}_i) \times \text{VBD_CNV}$$